Session 4A: Functions and Modules

Trainer Reference Guide

Purpose and outcomes

Extract repeated logic into small functions, distinguish printing from returning, use default/keyword arguments and scope, and organize reusable packages.

Examples and exercises

- 01_functions.ipynb builds a ‘full_name‘ function and guides a configurable department label. Its challenge separates capture, validation, and formatting.
- 02_scope_and_parameters.py builds email addresses with module-level defaults and local variables.
- 03_modules_packages.ipynb introduces standard/local imports and ‘__init__.py‘ interfaces.
- utils/validators.py and formatters.py provide reusable email/non-empty validation and employee formatting.
- exercise_01_email_function.py is guided validation. exercise_02_formatter.py independently returns a formatted line. challenge_registration_package.py imports both helpers into a registration workflow.

Trainer key points specific to this module

Ask ?What goes in, what comes out, and what side effect occurs?? Require functions that return reusable values. Demonstrate working-directory import problems without hiding them. Avoid premature abstraction: one purpose per helper is enough.

Portal connection and evidence

The portal divides routes, services, and repositories into modules. Learners can import helpers and test them with several arguments.

Delivery pattern

1. Explain the idea briefly and ask learners to predict a result.
2. Run the simple example, then connect it to an employee/user workflow.
3. Pair learners for the guided task; require an individual attempt next.
4. Use the challenge as the cumulative project increment.
5. Review the quiz and ask learners to explain, not only show, their code.

Trainer key points

- Keep live coding visible and deliberate. Normalize reading tracebacks.
- Ask what type and value each expression produces before running it.
- Let TODO files remain student-owned; open solutions only during review.
- Check boundary and failure cases, not only the happy path.
- End by tracing how the topic appears in the final portal.